

Corda

<https://www.corda.net/>

- Platform for blockchain and the distributed ledger
- Smart contracts
- Supports the development of distributed apps
- Notary infrastructure for sequencing and validation of transactions
- Flow based framework for negotiation and communication between the participants

Credits

<https://credits.com/>

- Smart contract programming
- Real-time monitoring of network transactions
- Web wallet with security using private and public keys

BigChainDB

<https://www.bigchaindb.com/>

- Big Data enabled blockchain database
- Decentralised management and control
- Dynamic management of digital assets
- Byzantine Fault Tolerance (BFT) for high performance computing applications
- Rich permissioning at each transaction
- Integration with MongoDB NoSQL for fast transaction processing with unstructured data
- Resistant to tampering; hence secure

Quorum

<https://www.jpmorgan.com/global/Quorum>

- Enterprise level smart contract and distributed ledger platform
- Peer permissioning

- High performance using Raft based consensus
- Exchange of private messages with secured contracts
- Fully customisable for large scale business and corporate applications
- Integration with CakeShop software development kit (SDK) for GUI enabled smart contracts, Quorum networks and APIs

There are other open source tools such as BigChainDB, Credits, Symbiont Assembly, Embark, Solidity, Truffle, etc, which interested readers can Google to learn more about.

Programming blockchain using open source development environments

Solidity

Solidity is a powerful, high-performance programming language for writing smart contracts. It follows the object-oriented programming paradigm with a higher degree of security and performance, which can be integrated with assorted blockchain platforms. The Solidity code is compiled and transformed to bytecode, which is executed on the Ethereum Virtual Machine (EVM). Solidity programming has the key base of many programming languages and scripts including Python, JavaScript and C++ so that it can be integrated with multiple environments and platforms for integration with blockchains. To work with Solidity, there are many integrated development environments (IDEs) and editors that can be used including Remix, EthFiddle and JetBrains.

To start with Solidity, use Remix, one of the powerful IDEs that is open source and also provides the Web based interface. Using the Web based interface of the Remix IDE, it is easy for developers to create smart contracts with blockchain programming.

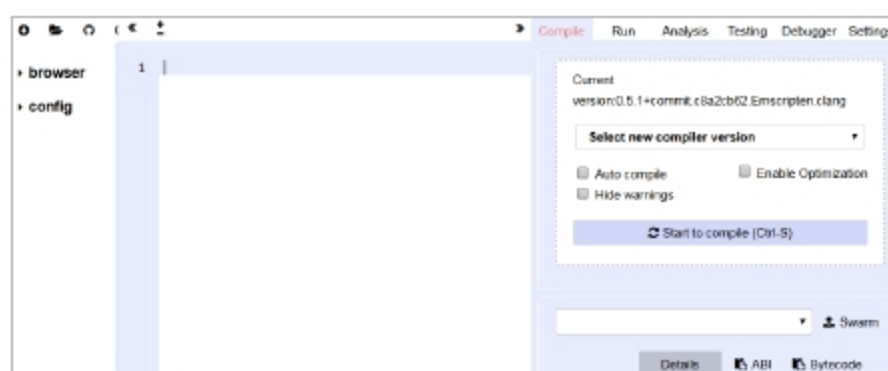


Figure 2: Remix IDE for blockchain programming



Figure 3: Writing a new currency in Solidity

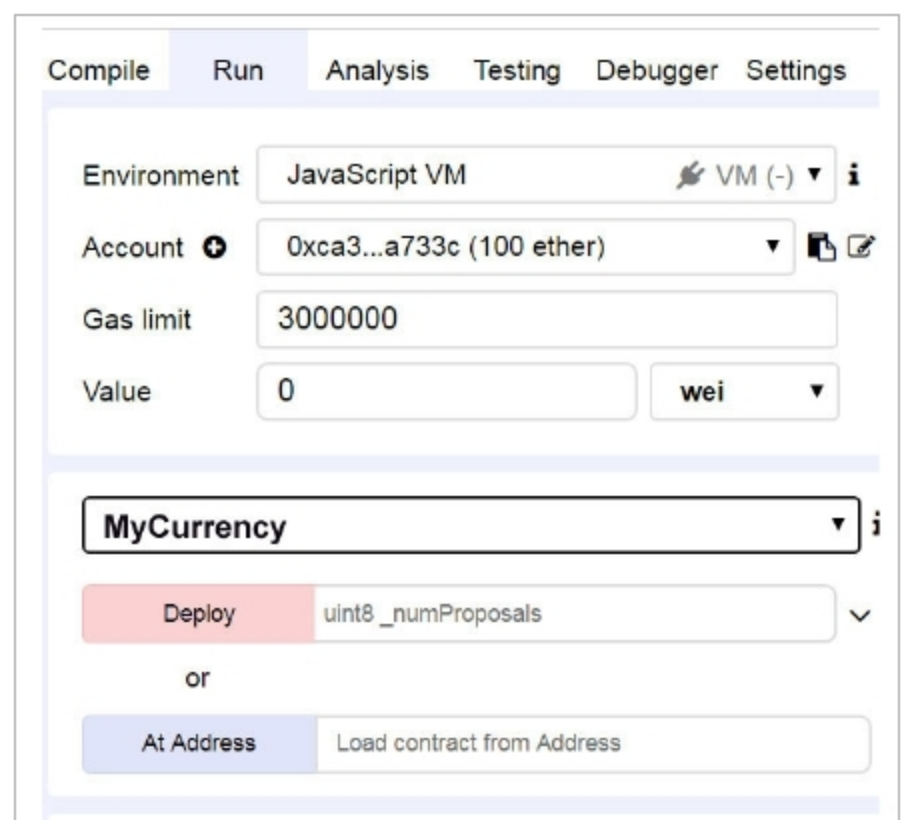


Figure 4: Compiling and running code in the Remix IDE